Design and Implementation of a Smart University ERP System with Role-Based Access Control and Academic Resource Management

CSA0904-Programming in Java

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Problem Statement

Data Silos & Security Risks

Manual campus management creates fragmented records and exposes sensitive data to unauthorized access.

No Centralized Access

Students, Faculty, and Admins lack a unified platform, leading to duplicated effort and miscommunication.

Inefficient Academic Operations

Grade processing, attendance tracking, and fee records are handled manually — slow, error-prone, and unsustainable.

Project Scope & Objectives

Secure Java Desktop ERP

Built with Swing/JavaFX and MySQL for a robust, cross-functional academic platform.

Role-Based Access Control

Distinct access privileges for Admin, Faculty, and Student roles enforced at every layer.

Full Academic Automation

Course registration, attendance, exams, fees, library, and reporting — all automated.

Literature Review

S.No	Author name	Year	Title of the paper	DrawBack	Overcome
1	Sharma, R., & Gupta, N.	2025	Role-Based Access Control in Higher Education Information Systems	Difficult management of permissions for large numbers of users.	Dynamic role assignment and centralized permission management.
2	Patel, D., & Mehta, S.	2025	Smart University ERP with Integrated Academic Resource Management	Limited analytics and reporting capabilities.	Include real-time dashboards, academic analytics, and performance reports for better decision-making.
3	Chen, X., & Wang, J.	2024	Intelligent Academic Resource Scheduling Using AI	Scheduling conflicts and inefficient classroom allocation.	Use automated scheduling algorithms to optimize academic resources.
4	Zhang, Y., & Li, H.	2024	Academic Resource Management System Using Web Technologies	Manual resource allocation leads to inefficient utilization.	Automate classroom, laboratory, and library resource allocation.
5	Ahmed, R., & Hussain, S.	2023	Secure Educational Management System Using Authentication Techniques	Weak authentication mechanism vulnerable to unauthorized access.	Apply secure authentication with RBAC and encrypted user credentials.

Technical Background & Tools

Core Language & Frameworks

Java (JDK 1 7+), Swing/JavaFX for GUI

Database & Data Access

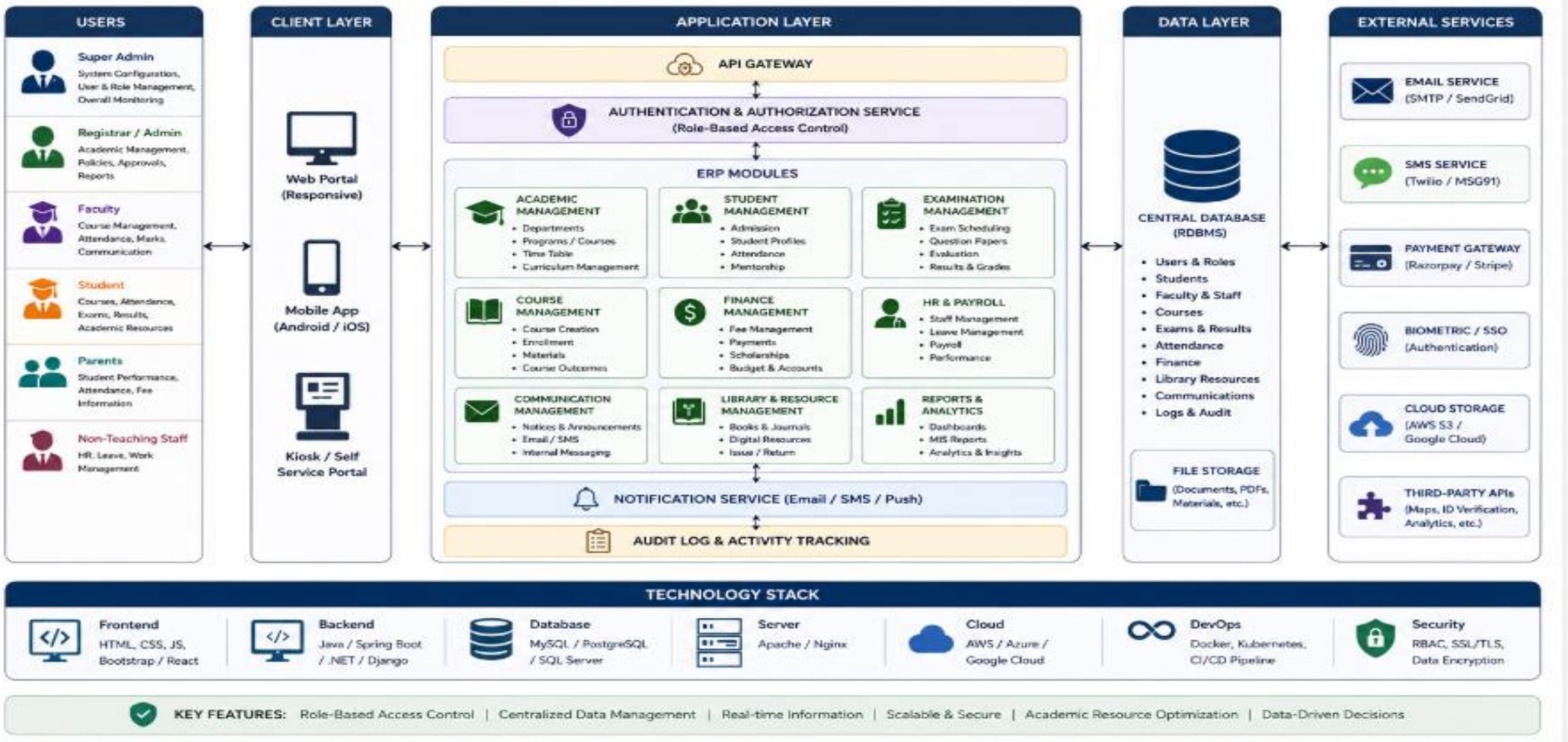
MySQL, JDBC API, Database Transactions

Programming Concepts

OOP, Collections Framework, File Handling, Exception Handling, BCrypt Password Hashing

Engineer to Excel

System Architecture



Key Project Modules

1. **Student & Faculty Management**

   Profile creation, role assignment, and academic record management.

2. **Course, Attendance & Examination**

   Course registration, real-time attendance tracking, and exam scheduling.

3. **Fee, Finance & Resource Management**

   Fee collection, financial transaction logging, and library resource tracking.

4. **Reports, Analytics & Admin Dashboard**

   Centralized dashboard with real-time analytics and exportable reports.

5. **Security, Audit Logs & Access Control**

   BCrypt-secured authentication, audit trail logging, and RBAC enforcement.

Evaluation Metrics & Testing

How We Measure Success

The system is validated across three critical dimensions to ensure it meets academic and enterprise-grade standards before deployment.

Zero transaction failures is a hard requirement for the database performance benchmark.

Security

Role authorization verification and BCrypt hashed password validation across all user types.

Database Performance

Query execution speed, JDBC connection pooling efficiency, and zero transaction failures.

System Usability

Functional CRUD operations verified and clean dashboard navigation tested end-to-end.

Expected Outcomes

Centralized Management

A unified academic and administrative management system eliminating data silos across all departments.

Secure Automation

Automated grade generation and attendance tracking with role-based security enforced at every step.

Real-Time Analytics

Efficient real-time analytics dashboards and comprehensive financial transaction logging for administrators.

Engineer to Excel